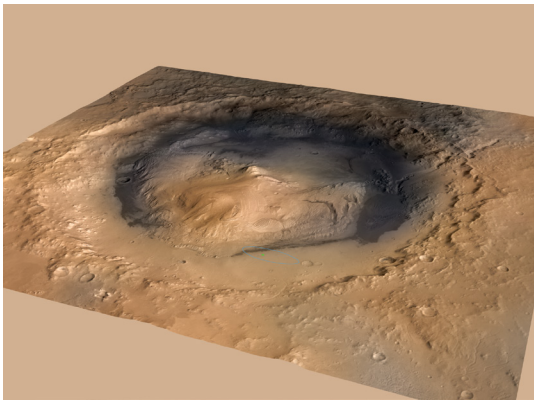
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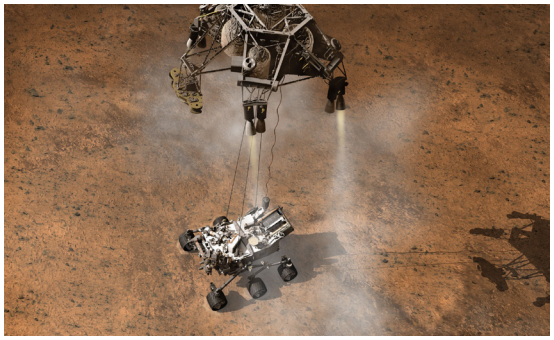
Mars Science Laboratory/Curiosity

NASA's Mars Science Laboratory mission set down a large, mobile laboratory — the rover Curiosity — at Gale Crater, using precision landing technology that made one of Mars' most intriguing regions a viable destination for the first time. Within the first eight months of a 23-month primary mission, Curiosity met its major objective of finding evidence of a past environment well suited to supporting microbial life. The rover studies the geology and environment of selected areas in the crater and analyzes samples drilled from rocks or scooped from the ground.

Curiosity carries the most advanced payload of scientific gear ever used on Mars' surface, a payload more than 10 times as massive as those of earlier Mars rovers. Its assignment: Investigate whether conditions have been favorable for microbial life and for preserving clues in the rocks about possible past life. More than 400 scientists from around the world participate in the science operations.



The rover's landing site, Gale Crater, is about the size of Connecticut and Rhode Island combined.



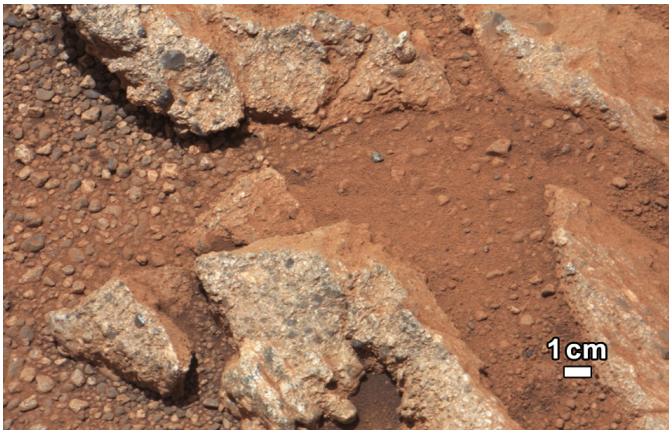
Curiosity touches down on Mars after being lowered by its Sky Crane.

Mission Overview

The Mars Science Laboratory spacecraft launched from Cape Canaveral Air Force Station, Florida, on Nov. 26, 2011. Mars rover Curiosity landed successfully on the floor of Gale Crater on Aug. 6, 2012 Universal Time (evening of Aug. 5, Pacific Time), at 4.6 degrees south latitude, 137.4 degrees east longitude and minus 4,501 meters (2.8 miles) elevation.

Engineers designed the spacecraft to steer itself during descent through Mars' atmosphere with a series of S-curve maneuvers similar to those used by astronauts piloting NASA space shuttles. During the three minutes before touchdown, the spacecraft slowed its descent with a parachute, then used retrorockets mounted around the rim of its upper stage. In the final seconds, the upper stage acted as a sky crane, lowering the upright rover on a tether to land on its wheels.

The touchdown site, Bradbury Landing, is near the foot of a layered mountain, Aeolis Mons ("Mount Sharp"). Selection of Gale Crater fol-



A rock outcrop called Link shows signs of being formed by the deposition of water.

lowed consideration of more than 30 Martian locations by more than 100 scientists participating in a series of open workshops. The selection process benefited from examining candidate sites with NASA's Mars Reconnaissance Orbiter and earlier orbiters, and from the rover mission's capability of landing within a target area only about 20 kilometers (12 miles) long. That precision, about a five-fold improvement on earlier Mars landings, made sites eligible that would otherwise be excluded for encompassing nearby unsuitable terrain. The Gale Crater landing site is so close to the crater wall and Mount Sharp that it would not have been considered safe if the mission were not using this improved precision.

Science findings began months before landing. Measurements that Curiosity made of natural radiation levels during the flight from Earth to Mars will help NASA design for astronaut safety on future human missions to Mars.

In the first few weeks after landing, images from the rover showed that Curiosity touched down right in an area where water once coursed vigorously over the surface. The evidence for stream flow was in rounded pebbles mixed with hardened sand in conglomerate rocks at and near the landing site. Analysis of Mars' atmospheric composition early in the mission provided evidence that the planet has lost much of its original atmosphere by a process favoring loss from the top of the atmosphere rather than interaction with the surface.

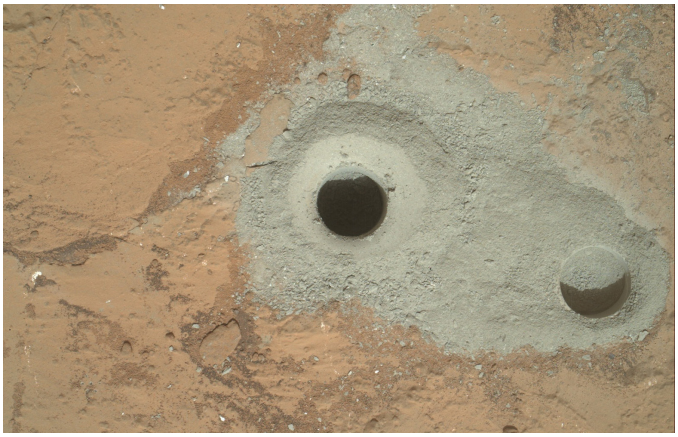
In the initial months of the surface mission, the rover team drove Curiosity eastward toward "Yellowknife Bay" to investigate an ancient river and fan system identified in orbital images.

The rover analyzed its first scoops of soil on the way to Yellowknife Bay. Once there, it collected the first samples of material ever drilled from rocks on Mars. Analysis of the first drilled sample, from a rock target called "John Klein," provided the evidence of conditions favorable for life in Mars' early history: geological and mineralogical evidence for sustained liquid water, other key elemental ingredients for life, a chemical energy source, and water not too acidic or too salty. On a subsequent drill sample, Curiosity was able to accomplish a first for measurements on another planet: determining the age of the rock. The measurements showed that the drilled material was 4.2 billion years old and yet had been exposed at the surface for only 80 million years.

In July 2013, Curiosity finished investigations in the Yellowknife Bay area and began a southwestward trek to the base of Mount Sharp. It reached the base layer of this main destination in September 2014. In the low layers of Mount Sharp during the rover's extended mission, researchers anticipate finding further evidence about habitable past environments and about the evolution of the Martian environment from a wetter past to a drier present.

Big Rover

Curiosity is about twice as long (about 3 meters or 10 feet) and five times as heavy as NASA's twin Mars Exploration Rovers, Spirit and Opportunity, launched in 2003. It inherited many design elements from them, including six-wheel



Curiosity's first sample drilling, at a rock called "John Klein."